

- 2 Antibiotic resistance of pathogenic bacteria is a worldwide cause for concern. Investigations into the use of antibiotics and their effects use an international unit of measurement, called the defined daily dose (DDD).

The DDD represents the assumed average dose for a drug per day being used to treat a specific disease in adults.

A study investigated the antibiotic resistance of *Streptococcus pneumoniae*. The study was carried out in some European countries and the percentage of *S. pneumoniae* resistant to the antibiotics penicillin and macrolide was determined.

The results are shown in Table 2.1.

**Table 2.1**

| country     | DDD per 1000 people per day |           | percentage of resistant <i>S. pneumoniae</i> |           |
|-------------|-----------------------------|-----------|----------------------------------------------|-----------|
|             | penicillin                  | macrolide | penicillin                                   | macrolide |
| Netherlands | 9                           | 1         | 1                                            | 5         |
| Denmark     | 11                          | 2         | 2                                            | 2         |
| Sweden      | 13                          | 1         | 4                                            | 3         |
| Germany     | 14                          | 2         | 7                                            | 9         |
| UK          | 14                          | 3         | 11                                           | 19        |
| Austria     | 18                          | 4         | 12                                           | 11        |
| Italy       | 24                          | 5         | 13                                           | 29        |
| Portugal    | 29                          | 4         | 29                                           | 9         |
| France      | 37                          | 6         | 43                                           | 53        |
| Spain       | 33                          | 6         | 50                                           | 36        |

- (a) (i) State **two** conclusions that can be made about the quantity of these antibiotics used and the percentage of *S. pneumoniae* resistant to these two types of antibiotic.

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- (ii) Calculate the ratio of penicillin use between Spain and the Netherlands.

ratio .....[1]

(b) To find the strength of the relationship between antibiotic use and the percentage resistance in *S. pneumoniae*, a linear correlation test was used. The probability for each value was found using the critical values from a probability table.

(i) State **two** reasons why this statistical test is suitable for these data.

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(ii) The results of the test are shown in Table 2.2.

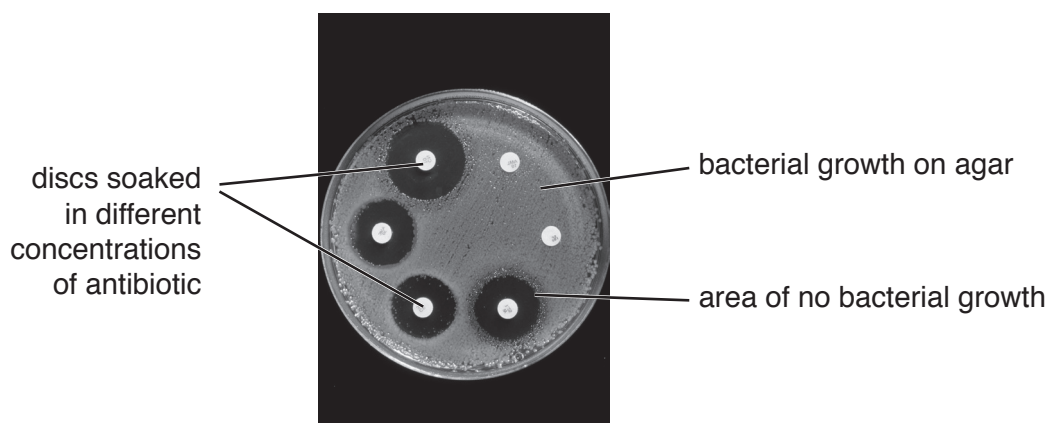
**Table 2.2**

| type of antibiotic | Pearson's linear correlation coefficient ( $r$ ) | level of significance ( $p$ ) |
|--------------------|--------------------------------------------------|-------------------------------|
| penicillin         | 0.95                                             | < 0.01                        |
| macrolide          | 0.86                                             | < 0.01                        |

State what the values in Table 2.2 indicate about the relationship between antibiotic use and antibiotic resistance in *S. pneumoniae*.

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- (c) The disc diffusion method is used to test the effectiveness of different concentrations of antibiotics. Fig. 2.1 shows the results of a disc diffusion test.



**Fig. 2.1**

Starting with a stock solution of an antibiotic, outline how you would make a serial dilution to produce a range of concentrations of the antibiotic.

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- (d) The disc diffusion test is also used to test for antibiotic resistance. The test may take about five days to complete.

*Escherichia coli* is a common bacterium in the human intestines that can develop antibiotic resistance and transmit it to other bacteria. It is present in faeces and can also be present in waste water that is discharged into rivers.

Environmental agencies test samples of waste water for antibiotic resistant bacteria. This can be done with the disc diffusion method. A new test was developed to allow estimates of antibiotic-resistant *E. coli* to be obtained within 24 hours.

Suggest **two** criteria that environmental agencies could apply when assessing the new test to see whether it is suitable.

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